Study Guide: Cellular Respiration

1. Overview

Cellular respiration is the process by which living cells break down organic molecules to produce adenosine triphosphate (ATP), the primary energy currency of the cell. In aerobic respiration, glucose and oxygen are converted to carbon dioxide, water, and energy through a series of tightly regulated enzymatic reactions. This process occurs in three main stages: glycolysis, the citric acid (Krebs) cycle, and oxidative phosphorylation via the electron transport chain.

Understanding cellular respiration is foundational to cell biology, physiology, and biochemistry. Disruptions to respiration underlie many diseases, including mitochondrial disorders, ischemic injury, and certain cancers. The overall reaction can be summarized as:

$$C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O + \sim 30\text{–}32\ ATP$$

2. The Three Stages

Glycolysis

Glycolysis takes place in the cytoplasm and does not require oxygen. One molecule of glucose (6 carbons) is split into two molecules of pyruvate (3 carbons each). The process yields a net gain of 2 ATP and 2 NADH per glucose molecule. Key enzymes include hexokinase, phosphofructokinase (PFK), and pyruvate kinase.

Citric Acid Cycle (Krebs Cycle)

In the mitochondrial matrix, pyruvate is first converted to acetyl-CoA, releasing one CO_2 and one NADH per pyruvate. Acetyl-CoA then enters the citric acid cycle, a series of eight reactions that regenerate oxaloacetate while releasing CO_2 and transferring high-energy electrons to the carrier molecules NADH and $FADH_2$. Per glucose molecule (two turns of the cycle), the Krebs cycle produces 2 ATP, 6 NADH, and 2 $FADH_2$.

Electron Transport Chain and Oxidative Phosphorylation

The electron transport chain (ETC) is located on the inner mitochondrial membrane. NADH and $FADH_2$ donate electrons to protein complexes (I–IV), releasing energy that pumps H^+ ions across the membrane, creating a proton gradient. ATP synthase harnesses this gradient to produce approximately 26–28 ATP per glucose. Oxygen serves as the final electron acceptor, forming water.

The stages are summarized in the table below:

Attribute	**Glycolysis**	**Krebs Cycle**	**Electron Transport Chain**
Location	Cytoplasm	Mitochondrial matrix	Inner mitochondrial membrane
Oxygen required?	No	No (indirectly)	Yes
ATP yield (per glucose)	2 net ATP	2 ATP	~26–28 ATP
Key inputs	Glucose, NAD^+, ADP	Acetyl-CoA, NAD^+, FAD	NADH, $FADH_2$, O_2
Key outputs	Pyruvate, NADH, ATP	CO_2, NADH, $FADH_2$, ATP	H_2O, ATP

3. Regulation and Efficiency

Cellular respiration is tightly regulated to match ATP production to cellular demand. Phosphofructokinase (PFK), the primary control point of glycolysis, is inhibited by high ATP concentrations and activated by AMP and ADP. Similarly, isocitrate dehydrogenase in the citric acid cycle is allosterically regulated by NADH and ATP.

The theoretical maximum yield of aerobic respiration is approximately 30–32 ATP per glucose molecule, though the actual yield in living cells is somewhat lower due to proton leakage across the mitochondrial membrane and the energy cost of transporting pyruvate and ADP into the mitochondria. Compared to anaerobic pathways (which yield only 2 ATP per glucose), aerobic respiration is dramatically more efficient.

- High AMP/ADP ratio → upregulates glycolysis and the citric acid cycle
- High NADH/NAD^+ ratio → inhibits key dehydrogenases
- High ATP concentration → inhibits PFK and other enzymes
- Oxygen availability → determines whether aerobic or anaerobic pathways are used

Key Terms

ATP	Adenosine triphosphate; the primary energy currency of the cell, produced during cellular respiration.
NADH / $FADH_2$	Electron carrier molecules that shuttle high-energy electrons to the electron transport chain.
Pyruvate	The 3-carbon end product of glycolysis, which is converted to acetyl-CoA in the mitochondrial matrix.
Acetyl-CoA	A 2-carbon acetyl group bound to coenzyme A; the entry molecule for the citric acid cycle.
Oxidative phosphorylation	The process by which the proton gradient generated by the ETC drives ATP synthesis via ATP synthase.
Substrate-level phosphorylation	Direct transfer of a phosphate group from a metabolic intermediate to ADP, producing ATP (occurs in glycolysis and the Krebs cycle).